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Arm Choices in a Sunburst Maze: Effects of Hippocampectomy in the Rat[1]

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HARLEY, C. W. *Arm choices in a sunburst maze: Effects of hippocampectomy in the rat.* PHYSIOL. BEHAV. 23(2) 283–290, 1979.—Hippocampectomized, neocortically damaged and unoperated rats were tested for response dispositions in a sunburst maze following training over a fixed path. The arm choices of neocortically damaged and unoperated rats were significantly limited to two of six alternatives. Hippocampectomized rats exhibited random selection of alternatives and were significantly different from controls. The neocortically damaged and unoperated rats were retested under scopolamine and retained a normal nonrandom selection pattern suggesting that normal hippocampal function was not necessary for the retention of these response selections. In accordance with this hypothesis, hippocampectomized rats trained preoperatively and tested for postoperative retention also exhibited a non-random selection pattern similar to controls, but they differed significantly in exploratory behavior. Choice patterns were stable over retests in the three groups. These data are consistent with the theory that the hippocampus is crucial in the acquisition of normal spatial dispositions, however spatial information once acquired may be available in the absence of hippocampus.

Hippocampectomy Scopolamine Spatial map Exploratory behavior

RECENTLY those interested in brain mechanisms have proposed that a primary function of the hippocampus is to generate spatial maps or to code stimuli in terms of spatial relationships. "In brief, the theory we constructed states that the hippocampus forms a map, that is, a pre-existent framework within which all stimuli are systematically related to each other forming what we have called a cartoon of that environment" [10].

This spatial mapping function for the hippocampus was first proposed in 1971 [12] following observations of hippocampal units whose firing was place specific. Since that first report, place coding by hippocampal units has been repeatedly confirmed [1, 11, 13]. These observations support the hypothesis that place information is available in the hippocampus but shed little light on how that information is developed and used by the animal.

The behavioral concept of a spatial mapping function was first elaborated by Tolman and his collaborators in the 1930s and 40s. Ritchie, working in Tolman's laboratory, argued in 1948 that the best evidence for a spatial mapping function was the existence of "auxiliary dispositions" [14] i.e., following training on a particular path rats acquire not only the single disposition to take the practised path or route, but auxiliary dispositions to take certain novel paths, previously untraversed, when the practised path was blocked. The development and retention of these auxiliary dispositions were used by Tolman and others as evidence [17] for the existence of cognitive maps.

In keeping with this position O'Keefe and Dostrovsky [12] had predicted that a rat deprived of the hippocampal map "could not learn to go from where it happened to be in the environment to a particular place independently of any particular route (as in Tolman's experiments)".

The task used by Tolman and his collaborators to demonstrate and probe auxiliary dispositions was a sunburst maze [18]. Rats were given repeated trials over a fixed and indirect pathway to a goal box. After training a test trial was given on which access to the fixed pathway was blocked and an array of alternate paths was available. In the original study [18] Tolman observed that 36% of his rats chose the alternate path leading most directly to the goal box. Other investigators [4,14] using the sunburst obtained evidence for other auxiliary dispositions. Typically there was a preference for one or two arms of the multichoice sunburst.

If the hippocampus is critical in generating spatial relational information, then one would predict that hippocampal damage would prevent the development of auxiliary dispositions during training in the sunburst maze. Further if the hippocampus is crucial for the storage of spatial relational information, one would anticipate a failure to retain normal auxiliary dispositions following hippocampal interference.

The present series of studies is concerned with: (1) investigating the auxiliary dispositions of normal rats on a modified sunburst task and (2) examining the consequences of hippocampal and neocortical disruption for the establishment and retention of normal auxiliary dispositions. As a corollary to these lesion studies, retention of auxiliary dispositions by controls is assessed under scopolamine.

METHOD

Animals and Surgery

Male Sprague-Dawley rats were obtained from Canadian Breeding Laboratories (Montreal, Quebec). For the testing

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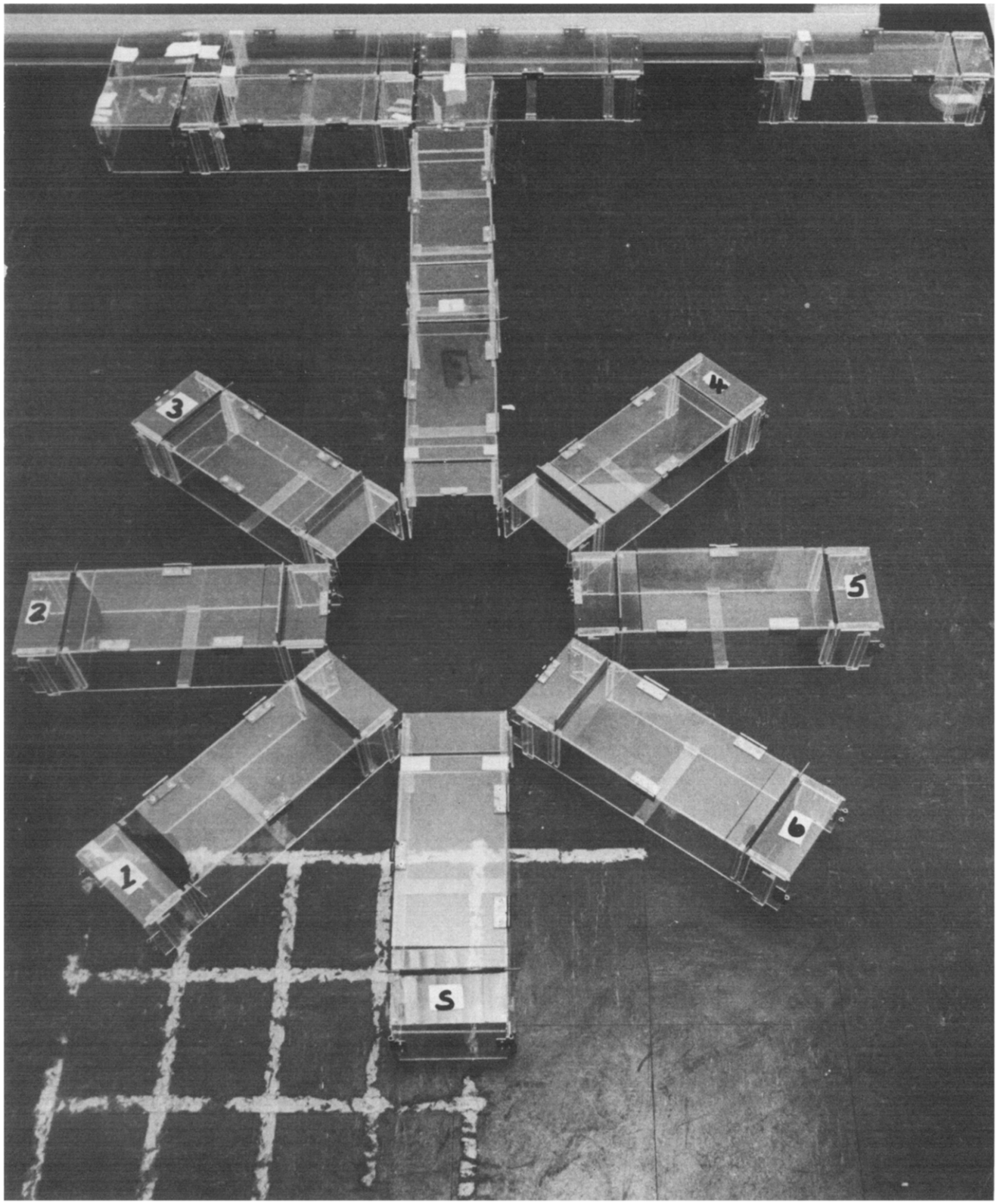


FIG. 1. Modified sunburst maze.

of acquisition of spatial dispositions in the sunburst maze 47 rats were used. For the testing of postoperative retention of spatial dispositions 36 rats were used. Rats of the acquisition group weighed from 220–300 g at the time of surgery. Rats of the retention group weighed 438–588 g at the time of surgery.

Surgeries were performed over a two week period. Rats were anesthetized with sodium pentobarbital (60 mg/kg). Removal of hippocampus and overlying neocortical tissue was performed in one stage using aspiration under a dissecting microscope. Control rats with only overlying neocortex removed were prepared alternately with hippocampectomized rats. In the acquisition group 16 rats underwent bilateral hippocampectomy, while fourteen rats underwent aspiration of the neocortex alone and 17 rats were unoperated. In the retention group, 12 rats received bilateral hippocampectomies while overlying neocortex alone was removed in 11 rats. Thirteen rats served as unoperated controls.

Apparatus

A modified sunburst maze was constructed of clear Plexiglas units (Fig. 1). Each Plexiglas unit was 13 cm wide, 10.5 cm high and 40 cm long with a hinged top. Eight of these units were fastened together at the corners with a cotterpin and hinge system to form a central arena approximately 35 cm in diameter. A clear sheet of Plexiglas was used to cover this open arena. All but two of the eight radiating maze units were blocked by masonite doors on the end nearest the center. The two open units were on opposite sides of the central arena. One unit served as the startbox, the other unit terminated in a black vinyl curtain, on the other side of the curtain an extended and indirect path constructed of modified units led to the goal box. The initial segment of the path leading from the arena was a total of 84 cm long, the segment after the first 90° turn was 54 cm, after the second 90° turn was 28 cm and after the third 90° turn leading to the goal box was 94 cm. The indirect path is illustrated in Fig. 1. A 25 cm gap between maze components (Fig. 1) was not opened until after an extended training period.

The apparatus was housed in a 3.05×2.40 m room. Kraft paper covered the floor area. The room had a single door and window opposite each other. The door opened near the start box and the window was 0.8 m above the goal box. A desk was present in one corner of the room and the experimenter occupied a chair near the door.

Procedure

During training, rats were maintained on 10 min of water daily. The research assistant running the training trials did not know the surgical assignment of the rats.

Acquisition group. The acquisition group was the first to be trained. For running convenience, the acquisition group of rats was divided into two subgroups. Subgroup A consisted of 16 rats and Subgroup B of 31 rats. Experimental and control rats were distributed over the two subgroups. Subgroup B was gentled and trained after Subgroup A. Prior to training rats received either one (A) or two (B) days of handling. Rats were placed in small groups on a table and stroked and handled for several minutes. The day following this brief gentling, training trials were initiated. A training trial consisted of putting a rat in the start box and permitting it to run the path to a 0.1% saccharin solution presented in a watch glass. Eight training trials were run each day for six days until 48 trials had been completed. All members of a subgroup were run on each trial. On the seventh day, two additional trials were given followed by a test for spatial dispositions in the sunburst.

The sunburst test consisted of blocking the normal entry to the indirect path and opening the six units which were normally closed. Each unit terminated in a black vinyl curtain. The test or exit choice was scored as the unit from which the rat exited through the black vinyl curtain, i.e., all four feet were outside the curtain. The rat was then picked up and returned to its home cage. No reward was available during the test.

Following the first sunburst test for the acquisition group each rat was given ten additional training trials on the indirect path using a 25 cm open space between the path and goal box. The gap training was added to give the rats experience with accessing the goal box from open space. If a rat refused to cross the open space on early trials it was forced across and permitted to obtain the reward. By the end of the ten trials all rats were obtaining the reward without forcing. The last two training trials were given on the same day and followed by a second sunburst test.

In Subgroup A three hippocampectomized rats and one neocortically damaged rat consistently refused to run during the initial training period and were excluded from the analysis of results. In Subgroup B three normal rats, three rats with neocortical damage, and three hippocampectomized rats balked on one or more training trials. All of these rats received at least 40 trials in the maze, however, and their choices were not different from other members of their treatment group. All Subgroup B rats were included in the analysis of results.

Acquisition controls tested under scopolamine. Four days after the second sunburst test for Subgroup B, all control rats (N=29) from the acquisition group which had chosen arms 3 and 4 in the first tests, were given 8 additional training trials in the single path maze with the 25 cm gap present. The following day an additional two training trials were given. After the tenth training trial each rat was weighed and assigned randomly to a scopolamine (1 mg/kg) or a saline (0.9% vehicle solution) injection. Drugs were administered intraperitoneally. Drug solutions and assignments were color coded and drug identity was not known to the experimenter.

Fourteen rats (five with neocortex damaged; nine unoperated) were assigned scopolamine injections. Fifteen rats (seven with neocortex damaged; eight unoperated) were assigned saline injections. Approximately thirty minutes after injection each rat was given a third test in the sunburst maze.

Retention group. The retention group of rats was trained at a later time than the acquisition group. Rats in this group were not trained on weekends and received a daily 20 min water period on the weekend. In addition, these rats were not run in subgroups, except on the first two trials which were run over a four day period with half the rats run each day. Rats were permitted five minutes to run down the path and obtain reward, which in this case was water, not 0.1% saccharin. If a rat did not run within the time limit, it was placed in the goal box and permitted to drink. By the third trial no placing was required. Backtracking was prevented in this group by closing the masonite doors at intervals along the path. After Trial two all rats were run one trial each day. During the first 28 trials there was no open space on the goal path. After Trial 28 the 25 cm space before the goal box was opened and the remaining trials were run with the open space. The first sunburst test (preoperative) was given after

TABLE 1

	Arms	Lesion: Unoperated 1 2 3 4 5 6	Neocortical 1 2 3 4 5 6	Hippocampal 1 2 3 4 5 6
Sunburst	1	0 0 5 12 0 0	0 1 4 8 0 0	1 2 3 2 5 0
Test	2	0 0 9 8 0 0	0 0 5 8 0 0	2 2 1 2 4 2

40 training trials as described previously. Besides recording the exit choice of each rat, the arm entries into the various units of the sunburst followed by a return to the central arena prior to an exit choice, were also noted. Additionally the latency to reach the goal box during training and to make a choice during testing were recorded.

Eight weeks after the preoperative test, 23 rats underwent the surgery described. In the interval prior to surgery the retention group rats were given 15 startle trials with a 110 dB stimulus at 30 sec interstimulus intervals. Rats were selected for surgery with the restriction that the startle levels of the hippocampectomized and control groups were similar. The startle matching was done for another experiment run after the maze testing.

Following a minimum two week recovery period all rats were retrained in the sunburst maze with the choice arms closed. On the first and second retraining days one trial was given. On the third and fourth day two trials were given. On the fifth day one training trial was followed by a first postoperative sunburst test. This single trial then test procedure was repeated on Days 6 and 7. Choices, latencies and prechoice entries were again recorded for each of the three postoperative tests.

All operated rats were eventually given an overdose of sodium pentobarbital and perfused with 0.9% saline followed by 10% Formalin. Brains were removed and sectioned at 40 microns on a freezing microtome. Every third section through the area of the lesion was taken on slides and stained with a 0.05% thionin solution.

RESULTS

Acquisition group. Sunburst Tests 1 and 2. On the first sunburst test all 17 unoperated rats chose either Arm 3 or Arm 4 as exit points (see Table 1). Arm 4 pointed directly at the goal box. Arm 3 pointed to the beginning of the final goal path. Twelve rats chose Arm 4; five chose Arm 3. As no particular pattern had been predicted the choice pattern exhibited was tested against a random exit distribution using the Kolmorogov-Smirnov test. The choice distribution of unoperated rats was significantly different from a random distribution $D=0.67$, $p<0.01$.

On the second sunburst test 53% of the unoperated rats changed their exit choice from Test 1. The change from Test 1 choices argues against exit choices being determined by approach to a single attractive stimulus. However, as in Test 1, all choices were confined to either Arms 3 or 4, hence there was no overall change in choice distribution. Eight rats chose Arm 4, nine rats chose Arm 3. This was also significantly unlike a random distribution ($D=0.67$, $p<0.01$).

Rats with neocortical damage exhibited a similar pattern. Twelve of the thirteen rats in this group chose either Arms 3 or 4 on the first sunburst test. Eight rats chose Arm 4, four rats chose Arm 3. On the second sunburst test all rats with neocortical damage chose either Arms 3 or 4 (see Table 1). Sixty-four percent of the rats with neocortical damage changed their choices from Test 1 to Test 2. These patterns were like those of the unoperated rats and were significantly different from a random distribution. (On Test 1, $D=0.67$, $p<0.01$ and on Test 2, $D=0.59$, $p<0.01$.)

TABLE 2

EXIT CHOICES OF ACQUISITION CONTROLS UNDER SCOPOLAMINE

Arms	1	2	3	4	5	6
Saline (n=15)	0	0	7	7	1	0
Scopolamine (n=14)	0	0	6	8	0	0

In contrast, the choices of the thirteen hippocampectomized rats were distributed over five of the six arms in Test 1 and over all of the six arms in Test 2. Neither choice distribution differed from a hypothetical random choice distribution. Variability between tests was more like that of controls with 60% of the hippocampectomized rats changing exit choices from Test 1 to Test 2.

Acquisition controls tested under scopolamine. The twenty-nine control rats (both unoperated and neocortically damaged) which had chosen Arms 3 and 4 in the previous sunburst tests were assigned to either scopolamine or saline injections and tested a third time. Rats receiving scopolamine again chose either Arm 3 (n=6) or Arm 4 (n=8). This choice pattern was unlike that of a random distribution ($D=0.67$, $p<0.01$). Fourteen of the rats receiving saline also chose either Arm 3 (n=7) or Arm 4 (n=7). One rat with neocortical damage chose Arm 5. This choice pattern was unlike that of a random distribution ($D=0.60$, $p<0.01$) (see Table 2).

Six rats under scopolamine chose an arm different from that chosen on Test 2 while 8 rats under saline switched choices from Test 2. Of the six scopolamine rats three switched for the first time on this test, while in the saline group two switched for the first time in this third test.

Retention group. Preoperative and postoperative exit choices of the retention group are presented in Table 3. Given the spatial patterns of control rats in the acquisition group (see Table 1) and the preoperative choices of the retention group (see Table 3), the six choice categories were reduced to two for purposes of analysis. Instances of choosing Arms 3 and 4 were compared to choices of all other arms. A random choice model predicts that Arms 3 or 4 would be chosen with a probability of 1/3, while the probability of choosing other than Arms 3 or 4 is 2/3. Using the binomial expansion theorem it is possible to compute the exact prob-

TABLE 3

PREOPERATIVE AND POSTOPERATIVE TEST CHOICES OF THE RETENTION GROUPS

Test	Unoperated 1 2 3 4 5 6	Lesion Neocortical 1 2 3 4 5 6	Hippocampal 1 2 3 4 5 6
Preop	0 0 5 6 1 1‡	0 0 8 3 0 0‡	0 0 6 6 0 0†
Postop 1	0 0 5 6 1 1‡	0 0 3 6 1 1‡	1 0 6 2 2 1*
Postop 2	0 0 8 5 0 0‡	1 1 7 2 0 0‡	1 2 3 5 0 1†
Postop 3	0 0 7 4 1 1‡	0 0 8 2 0 1‡	0 1 5 4 1 1‡

*$p<0.02$.
†$p<0.005$.
‡$p<0.001$.

ability of a particular distribution. If the binomial expansion model is applied to the acquisition group data the pattern of significant differences does not alter. In the retention group however both control rats and the hippocampectomized rats differed significantly from the random choice model on all tests both pre- and postoperatively (see Table 3). The probability of a given two choice distribution or one more extreme is indicated in Table 3.

The flexibility of choice over the four tests is reflected in the number of different exit choices made out of the total possible of four exits. The mean for all groups was slightly over two differing exit choices on the 4 tests. No more than two rats in any group failed to alter their choice at least once. Of those control rats which varied their exit choices over testing more than 85% included both Arm 3 and Arm 4 among their choices. In the hippocampectomized group 60% chose both Arm 3 and Arm 4. For comparison with the acquisition group data the percentage of rats in each group changing choices in any given pair of the four sequential tests is shown in Table 4. Note that the delay between Preop Test and Postop Test 1 is nearly twelve weeks and variability among the unoperated rats is lower over this pair of tests. In general, however, experimental and control rats were similar in altering their choices over tests in both the acquisition and retention groups.

Arm entries prior to an exit choice provided a different picture of flexibility in the retention group. For this measure an entry into a choice arm followed by a return to the central arena was counted as one arm entry. The exit arm choice was also scored as one arm entry. Minimum entry score is thus one. Table 5 presents the mean number of arm entries for all retention group tests. A Lesion (3) × Test (4) ANOVA on the number of arm entries revealed a significant lesion by test interaction, $F(6,99)=2.68$, $p<0.05$.

Tests of simple effects indicated no differences among the groups on the Preop Test. The overall mean arm entries were 2.4. In contrast on Postop Tests 2 and 3 both control groups had substantially increased in number of arm entries with a combined mean=12.81 arms and differed significantly from the hippocampectomized rats, combined mean=2.95 arms (Postop Test 2, $F(2,36)=12.9$, $p<0.01$; Postop Test 3, $F(2,36)=5.93$, $p<0.01$). On Postop Test 1 there was also a significant difference among the groups (Test 1, $F(2,36)=3.92$, $p<0.05$) but this is attributable primarily to differences between hippocampectomized rats, $\bar{X}=1.33$, and unoperated rats, $\bar{X}=9.30$. Neocortically operated rats were intermediate, $\bar{X}=4.64$.

Over the four tests then there was no significant change in the mean arm entries for the hippocampectomized rats while both neocortically damaged and unoperated rats showed marked increases in arm entries from the Preop Test to the Postop Tests with the largest change seen on Postop Test 2.

This increased number of arm entries appeared to reflect increased arm exploration. That is, among the control rats a majority were entering at least five of the six possible arms one or more times prior to an exit. On Postop Tests 2 and 3,

TABLE 4

PERCENTAGE OF RATS CHANGING EXIT CHOICES OVER TWO TESTS

	Acquisition Test 1 to Test 2		Retention Preop-Postop 1		Retention Postop 1-Postop 2		Retention Postop 2-Postop 3	
Unoperated	53%	n=17	31%	n=13	62%	n=13	62%	n=13
Neocortical	64%	n=13	73%	n=11	55%	n=11	55%	n=11
Hippocampal	60%	n=13	67%	n=12	75%	n=12	67%	n=12

TABLE 5

MEAN NUMBER OF ARM ENTRIES PRIOR TO AN EXIT CHOICE FOR ALL TESTS OF THE RETENTION GROUP

Lesion	Test	Preop	Postop 1	Postop 2	Postop 3
Hippocampal		1.75	1.33	1.58	2.92
Neocortical		2.75	4.65	13.91	12.00
Unoperated		2.69	9.38	14.38	11.00



FIG. 2. Photomicrographs of frontal sections from rat brains in the acquisition group with representative damage to hippocampus, (left side; S30), and to overlying cortex (right side; S33).

77% of the unoperated rats entered at least five different arms prior to an exit choice. Similarly among the neocortically damaged rats 64% entered five arms or more in Postop Test 2, while 82% entered at least five different arms on Postop Test 3.



FIG. 3. Photomicrograph of anterior damage in a neocortically lesioned brain from the retention group.

None of the hippocampectomized rats entered five or more arms on Postop Test 2 and only 12% did so on Postop Test 3.

As would be expected the greater number of arm entries postoperatively meant that exit latencies were longer for control rats than hippocampectomized rats. The correlation between latencies less than 30 sec and a single prechoice arm entry was .81.

Latencies were also assessed on all training trials. Only the training trials immediately preceding a test were examined statistically. The preoperative latency for entering the goal box before the first test was 18 seconds. The postoperative latencies were an average of 74 seconds for unoperated rats, 115 seconds for neocortically damaged, and 38 seconds for hippocampectomized rats on the three pretest runs. There were no significant differences among groups pre- or postoperatively.

Histology

Figure 2 illustrates the typical extent and nature of hippocampal and neocortical damage in a hippocampectomized brain (left) and a neocortically lesioned brain (right) of the acquisition group. In all hippocampectomized rats the dorsal hippocampus was completely transected bilaterally. Anterior remnants were seen in some brains (Fig. 2) but removal of the hippocampal tissue above the rhinal fissure was quite complete in all posterior sections. In the acquisition group ventral hippocampal tissue at and below the rhinal fissure was typically not removed. In the retention group damage to this ventral hippocampal segment was more characteristic. Also in the retention group removal of neocortex among the neocortically operated rats was more extensive at anterior levels than in Fig. 2. An example of this anterior neocortical damage is seen in Fig. 3.

Among the hippocampectomized rats extrahippocampal damage occurred unilaterally in septum (n=3), thalamus (n=10) and amygdala (n=2) in the two groups taken together. Damage in thalamus usually occurred in the lateral thalamic or lateral geniculate nucleus. In one hippocampectomized brain from the retention group bilateral thalamic damage was seen together with unilateral damage to superior colliculus and habenula. The removal of hippocampus was nearly complete in this rat and its data did not differ from that of other hippocampectomized rats. Its data were included in data analysis.

In the neocortically damaged brains unilateral damage to the dorsal hippocampal surface (extending to the level of the dorsal pyramidal cells) occurred in some sections in 16 brains. Bilateral surface damage occurred in 3 brains. Choices did not differentiate these rats from those without bilateral disruption.

DISCUSSION

The Sunburst Test

The modified sunburst test appears to elicit strong auxiliary or spatial dispositions in unoperated rats and in those with posterior neocortical lesions. Both groups select Arms 3 and 4 of the six choices available. To date, in our laboratory, 126 control rats have been tested in the sunburst with more than 90% choosing either Arm 3 or 4 on the first test. If repeated testing is included (rats have received as many as 4 probe tests) the total tests to date is 314 with again 90% of the choices to Arms 3 or 4. A particularly interesting outcome in the present experiment was the observation of flexible choice patterns among the control rats. In earlier studies [5, 14, 18] only single probe tests were given. In the present study approximately half the rats in any given control group altered their choices from one test to the next, but remained constrained to Arms 3 or 4. This constraint persisted despite the fact that control rats markedly increased their entries into other arms after nonreward testing. Yet they did not exit from these other arms. The constrained but flexible response patterns argue for a response mode more characteristics of problem solving with maps than by strict S-R associations [10].

It seems the arm choices exhibited in the sunburst are held in common by most experienced rats and have considerable stability. It is not possible, however, within the modified sunburst to know if the basis for these choices is: (a) the relationship of the exit arms to the goal box or final goal path, or (b) the relationship of the exit arms to the initial fixed arm segment. In the original maze Tolman found 36% of the rats tested selecting an arm pointing to the goal box [18] while Gentry, unable to replicate Tolman, found 52% of the rats tested in another sunburst choosing arms adjacent to the initial path [5]. Both experimenters used considerably fewer trials than in the present experiment. Ritchie, in a somewhat different test, found clear and consistent spatial choices, but was unable to identify the determinants as a shortcut tendency, hence the term "auxiliary dispositions"[14].

Whether normal arm choices are determined by location of the fixed path or goal arm location, the random choices of hippocampectomized rats in the sunburst, particularly as contrasted with the neocortically damaged rats, argues for the unique inability of hippocampectomized rats to generate normal responses vis-a-vis spatial location [10].

Postoperative Retention

On the other hand, the present data argue for the availability of spatial dispositions, once acquired, from extrahippocampal structures. Under scopolamine, an anticholinergic which should interfere with the acetylcholine system projecting from septum to hippocampus and which has been reported to mimic the effects of hippocampectomy on some tests [2], rats were not impaired in the retention of previously acquired spatial dispositions. Moreover, the choices of the drugged rats appeared as flexible as those of saline controls, i.e., choices on this third test were not simple perseverations of Test 2 choices. More conclusively, with rats trained preoperatively, no significant disruption of acquired spatial dispositions occurred following radical hippocampectomy. The normal dispositions of these hippocampectomized rats argues strongly that relational information is available in the absence of the hippocampus if acquired preoperatively. This is a problem of considerable interest and deserves further evaluation. In the human literature anecdotal reports would appear to support the availability of preoperatively acquired spatial information following hippocampal damage, e.g., H.M. is reported to have directed psychologists to his old home, but was unable to similarly locate his new home [9].

In the animal literature there are no relevant preop/postop studies with hippocampal damage. Recent papers by Sinnamon *et al.* [15] and Jarrard *et al.* [7] looking at preop and postop performance differences on spatial tasks use tasks which depend on short term spatial memory. Neither investigator is assessing long-term retention of spatial information. In earlier studies Kaada *et al.* [8] and Thomas [16] had looked at complex spatial maze retention in preoperatively trained rats and found that hippocampal damage resulted in postoperative deficits. Kaada, however, intentionally used rats which had not been trained to criterion preoperatively and argued that criterion-trained rats would probably not have been affected postoperatively. Thomas' study is more clearcut in identifying a postoperative relearning deficit in a four unit U maze. However it is unlikely that Thomas' maze could have been solved by any simple hypothesis about spatial relationships. Hence as argued at the outset, there are no studies which directly assess spatial relational storage postoperatively. Of the experiments available, only Thomas' raises the possibility that long-term storage is dependent on hippocampus. The present data argue strongly to the contrary. The retention of auxiliary dispositions in the sunburst task does not appear dependent on an intact hippocampus.

Exploration

A third outcome of these studies was the striking disparity in exploratory activity following hippocampal lesions. Nadel and O'Keefe [10] have argued that hippocampectomized rats, while often seen as more active in settings which elicit normal exploration, e.g., open field, are not in fact exploring. Their indistractability in other settings [3,6] is consistent with this view and the failure of disconfirmation of expectation, i.e., nonreward, to trigger arm entries in the sunburst would appear to extend and confirm their hypothesis.

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